

brown open silicon

designing open chips. together.

meet the team



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what is **chip design**?

chip design (*noun*)

the process of specifying, modeling, and creating the architecture and physical layout of an integrated circuit (IC), including its logic, timing, power, and manufacturing constraints, using specialized hardware description languages and electronic design automation (EDA) tools.

overview: **BLOC**



build

tapeout
reproducible
research

learn

VLSI and chip
design
project
management skills

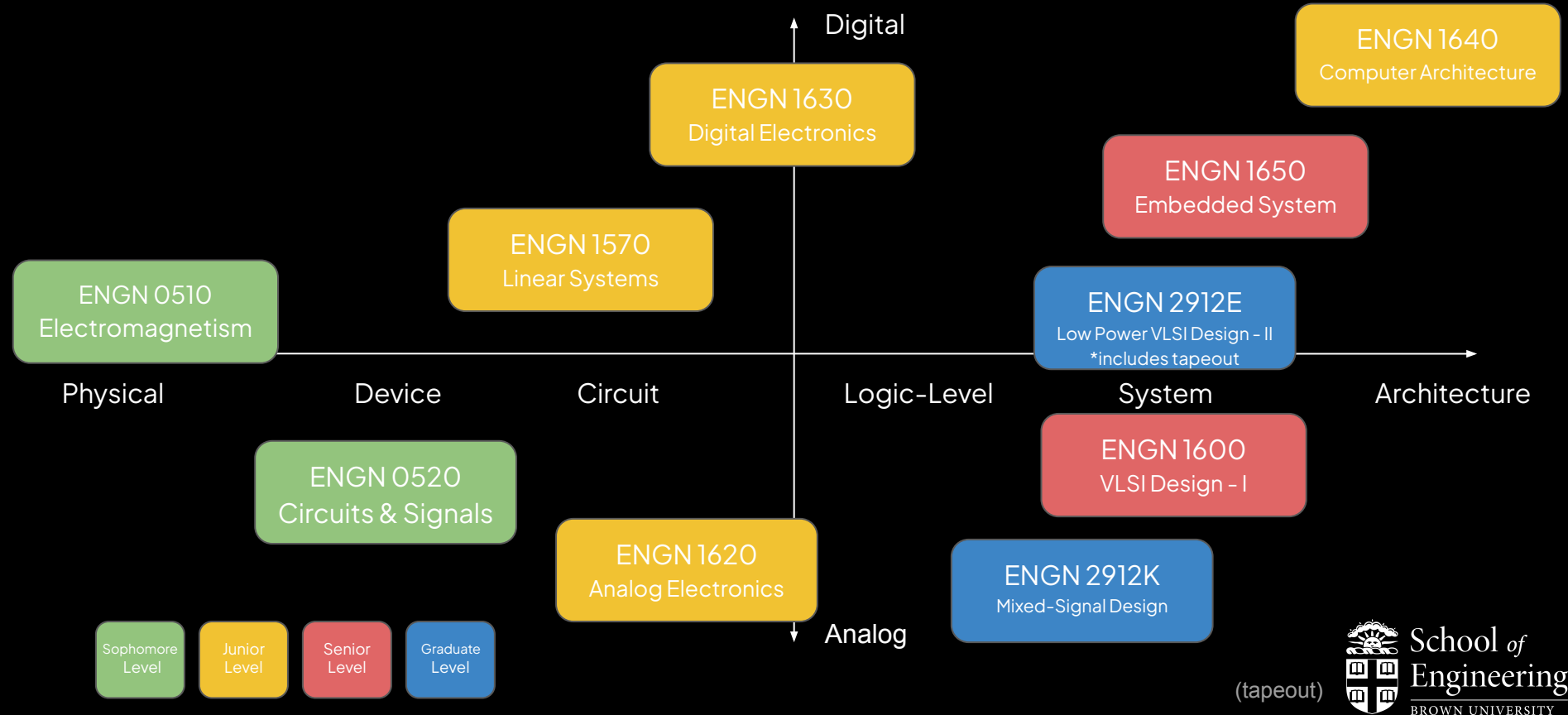
open

open source tools
for chip design
open to all
students

connect

industry leaders
leading
researchers
peers and faculty

ECE classes at brown



overview: opportunities



workshops

introduction to VLSI and chip
design

Introduction to chip design
tools

research

research opportunities
with faculty

paper publications

travel grants

IEEE SSCS
Code-a-Chip

overview: opportunities



ISSCC 26 Code-A-Chip Travel Grant Winners:

1. CircuitsDNA: Evolutionary Synthesis of Approximate Hardware acceleration

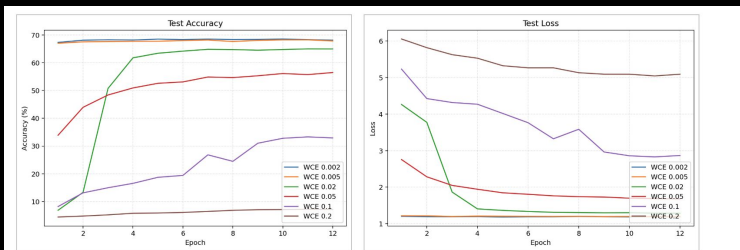
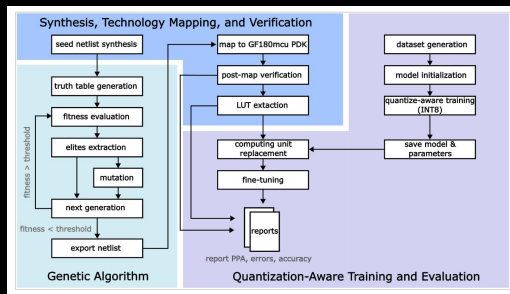
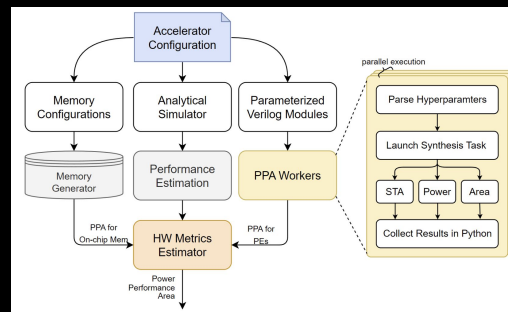
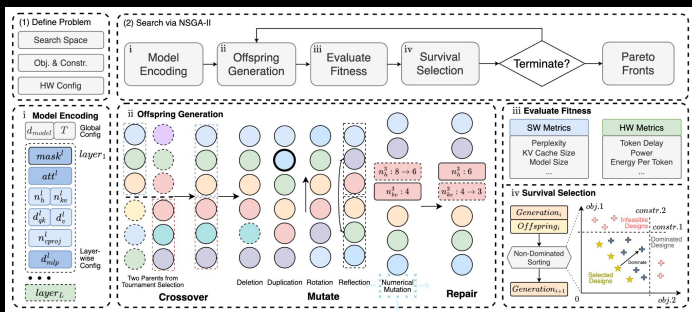


Figure 9: Test accuracy and loss during hardware-aware fine-tuning with approximate multipliers of different relative WCE values.

2. LLMForge: Joint HW/SW Optimization of Edge-Scale Transformer Architecture



what makes **you** interested
in **chip design**?

Potential **BOS** Projects



AI for Chip Design

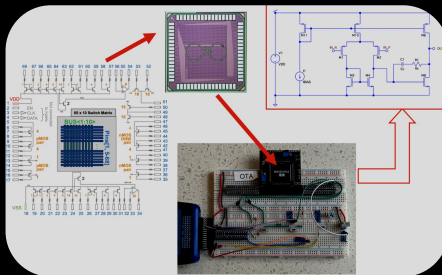
gLayout – an LLM driven framework for layout automation

OpenFASOC – synthesizable circuits generation

OpenROAD, circuitsDNA, LLMForge

MOSbius style projects

transistor-level analog design to build chips capable of being used on a breadboard



Other Ideas?

Technical Committee on OSE initiative



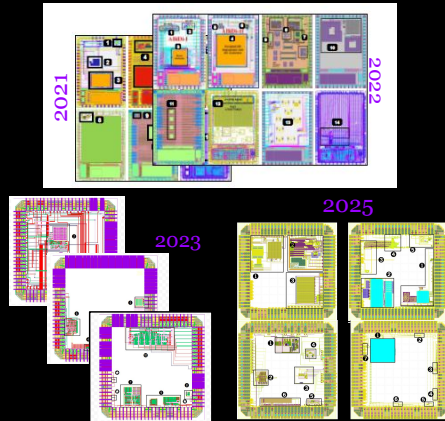
- ❖ Running PICO Chipathon 2021-'25
- ❖ Running “Code-a-Chip” travel grant competitions for
 - VLSI 2023-'26
 - ISSCC 2023-'25
- ❖ Contributing to magazine articles “PICO Chronicles”
- ❖ Organized two magazine special issues
 - Design & Test magazine
 - SSCS magazine
- ❖ Workshop at VLSI 2025
- ❖ PICO TC-OSE Booth at ISSCC 2024-25

SSCS PICO – Core Activities Overview

SSCS PICO – Platform for Integrated Circuit Design Outreach

A Scalable Platform for Integrated Circuit Design Outreach enabling hands-on chip design experience for early-career of students globally

Annual "Chipathons"



CHIPS Alliance: Validated our mission and impact with over \$30,000 in financial donations to support our activities.

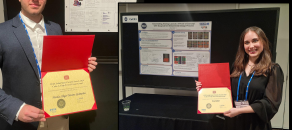
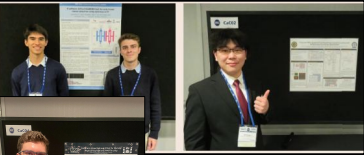
GlobalFoundries (GF): Provided key in-kind support with donations of free silicon, a critical resource for hands-on design.

IEEE SSCS: Our foundational partner, providing consistent and significant financial backing for the platform and its events.

SSCS PICO – Core Activities Overview

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"Code-a-Chip" Events

High-impact deployment. We take PICO to premier industry venues like ISSCC and VLSI, demonstrating platform value to top-tier academic and industry audiences.

PICO Booth @ ISSCC

Direct community engagement and a showcase for student success, attracting vital industry and academic attention to our mission and talent pool.



Open-Source Chipathons since 2021

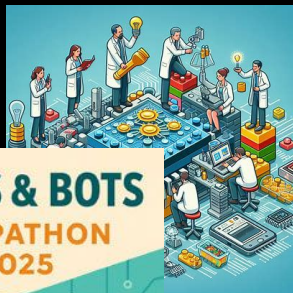


2022 SSCS "PICO"
Open-Source Chipathon
Proposal Deadline: May 1, 2022

**Your Design
Goes Here!**

2024 Chipathon
Automating Analog Layout

**Blocks & Bots: An
Open Chip
Playground
augmented with
LLMs**

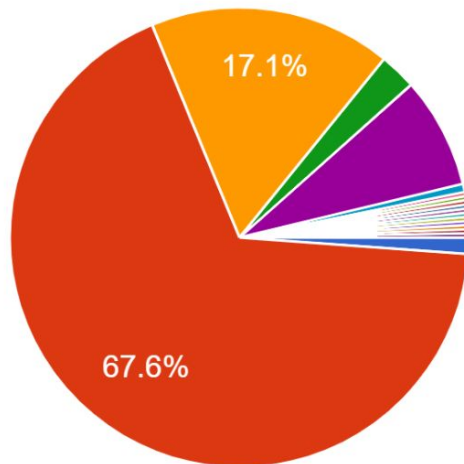


- **2021:** 61 submissions, 18 selected (11 taped out)
- **2022:** 54 submissions, 22 selected (14 taped out)
- **2023:** 10 selected (3 taped out)
- **2024:** Postponed
- **2025:** selected (4 taped out)

Chipathon 2025 - Early Infographics

What is your affiliation:

346 responses



- Academia (pre-college)
- Academia (undergraduate student)
- Academia (graduate student)
- Academia (postdoc or professor)
- Industry
- Affiliated to none, individual participant
- ISHI-KAI
- Hobbyist with eee background

2025 Chipathon: New Introductions



- ❖ Introduced **MOSbius** – a Playground for transistor level experimentations
- ❖ Introduced a **Digital Design** track to enhance existing digital libraries
- ❖ Introduced Prizes:
 - Certificate for Tape out
 - Certificate for testing your Chips
 - Paths towards Code a Chip Travel Grant
- ❖ Introduced Introductory Challenges for each track

Impact & Recognition through Workshops at Key Conferences

Open Source PDKs and EDAs, Community Experiences toward Democratization of Chip Design

Organizer : Makoto Ikeda (The University of Tokyo)

Organizer : Mehdi Saligane (University of Michigan)



What is Possible with Open Chip Design? The Journey so Far.

Organizer : Mehdi Saligane (Brown University)

Makoto Ikeda (University of Tokyo)

SSCS Magazine Special Issue

Tiny Tapeout

A shared silicon tape out platform accessible to everyone

Introduction
Open source tools and proven design methodologies are making it easier for designers to create custom silicon. However, the high cost of silicon tape outs remains a barrier for many small and medium-sized enterprises. The Tiny Tapeout project aims to lower this barrier by providing a shared silicon tape out platform accessible to everyone.

As the tools you can create (read out) are more powerful, the silicon you can create is more powerful. The Tiny Tapeout project is a shared silicon tape out platform accessible to everyone. The project is a shared silicon tape out platform accessible to everyone. The project is a shared silicon tape out platform accessible to everyone.

Rapid Prototyping of Laser-Induced Graphene Sensors With Open-Source Silicon

Paving the Way for Low-Cost and Robust Flexible Wearable Sensing

Introduction
The rapid prototyping of laser-induced graphene (LIG) sensors with open-source silicon is a promising approach for low-cost and robust flexible wearable sensing. This article describes the process of rapid prototyping of LIG sensors with open-source silicon, from design to fabrication and testing.

Unlocking Circuits for Quantum With Open Source Silicon

A first look at measured open source silicon results at 4 K

Introduction
The development of open source silicon for quantum circuits is a key challenge in the field. This article presents the first look at measured open source silicon results at 4 K, showing the potential for unlocking quantum circuits with open source silicon.

Open Se Cura

First silicon results of an auditable and transparent hardware root-of-trust system using open electronic design automation in 16 nm

Introduction
The development of an auditable and transparent hardware root-of-trust system using open electronic design automation (EDA) is a key challenge in the field. This article presents the first silicon results of such a system, showing the potential for unlocking quantum circuits with open source silicon.

Low-Cost Open Source ASIC Design and Fabrication

Creating your own chips with open source software and multiproject wafers

Introduction
The development of low-cost open source ASIC design and fabrication is a key challenge in the field. This article presents the process of creating your own chips with open source software and multiproject wafers, from design to fabrication and testing.

On the Versatility of the IHP BiCMOS Open Source and Manufacturable PDK

A step towards the future where anybody can design and build a chip

Introduction
The development of a versatile open source and manufacturable PDK for BiCMOS technology is a key challenge in the field. This article presents the process of creating your own chips with open source software and multiproject wafers, from design to fabrication and testing.

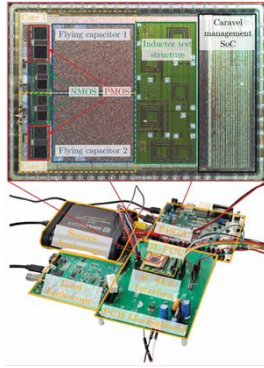
Open-Source ICs: Create your own recipe!

Introduction
The development of open source ICs is a key challenge in the field. This article presents the process of creating your own recipe for open source ICs, from design to fabrication and testing.



IEEE Design & Test – Special Issue on Open-Source Silicon

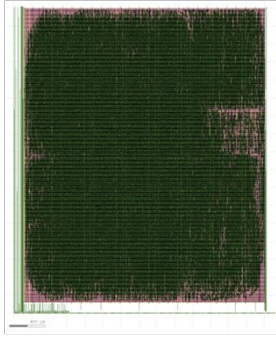
Open-Source Multilevel Converter Power IC Design and Test



Power Management
(Measured)

- ✓ Open-source PDK (SkyWater 130nm)
- ✓ Open-source tools
- ✓ Open-sourced design (Github)

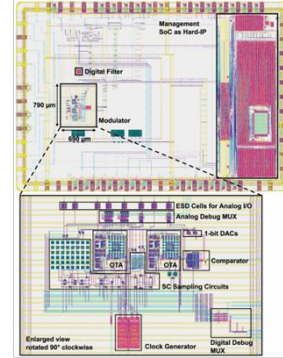
Leveraging Generative AI for Rapid Design and Verification of a Vector Processor SoC



Digital Processor
(Simulated, fabrication ongoing)

- ✓ Open-source PDK (SkyWater 130nm)
- ✓ Open-source tools
- ✓ Open-sourced design (Github)

An Open-Source 12-bit 10-kS/s Incremental ADC in 130-nm CMOS



ADC
(Measured)

- ✓ Open-source PDK (SkyWater 130nm)
- ✓ Open-source tools

SSCS Magazine – PICO Chronicles

Democratizing IC Design: The Story of a New Movement and the Launch of the SSCS PICO Program

In recent decades, the software community has greatly benefited from open source tools and code. It has embraced a highly productive culture of sharing and open collaboration that resonates with the spirit of our times. We live in an age of global

notes that the key to the success of open source in software is that “it depends on the ability of programmers to freely share their program code so that others can improve on it.” Bob Young, founder of Red Hat, notes in the foreword of the same

An open source PDK enables the broadest possible collaboration on any given design or across designs. That, in turn, increases the number of potential designers, which reduces the cost per design and makes it affordable. In a traditional model, teams are limited

SSCS PICO Chronicles: The Open Source Community is Growing Further and Supported by SSCS PICO

The previous year was another great success for the open source integrated circuit (IC) design ecosystem. It finished with two actions: the second IEEE Solid-State Circuits Society (SSCS)

conversations on the bustling IEEE-SSCS-DC-22 Slack channel.

In September 2022, the SSCS launched its first “Code-a-Chip” travel

ners attended the 2023 International Solid-State Circuits Conference, where they have been recognized for their awards during the Student Research Paper Competition (SRPC).

SSCS PICO Chronicles: News From the Open Source Community

Five Winners of SSCS Open Source Ecosystem “Code-a-Chip” Travel Grant Awards at VLSI 2024!

The IEEE Solid-State Circuits Society (SSCS) is propelling open source chip design to new heights with its successful “Code-a-Chip” Travel Grant Awards program. VLSI 2024 provided a platform to acknowledge the signif-

TABLE 1. CODE-A-CHIP TRAVEL GRANT WINNERS.

NAME OF TEAM LEADER	AFFILIATION	NOTEBOOK TITLE
Qi Tian	National ASIC Center, School of Integrate Circuits, Southeast University	ChatChisel: Enabling Agile Hardware Design with Large Language Models
Kristian	Universidad Industrial de	OBS Demo: A demonstration of

SSCS PICO Chronicles: News From the Open Source Community

Sixteen Teams Launched for 2024 SSCS PICO Open Source Chipathon and Exploring Analog IC Design Automation

The IEEE Solid-State Circuits Society (SSCS) has been holding its fourth open source integrated circuit (IC) design contest under the umbrella of its PICO (Platform for IC Design Outreach, <https://sscs.ieee.org/technical-committees/tc-ose/sscs-pico-program/>) program from 2024 May.

This year’s event was a broad call for examples of automated analog IC design using a combination of open source tools and open MPW. Of the many teams who submitted a 16 teams have moved on to

Figure 1 shows the analog circuits that the 16 teams are working on to achieve automatic generation. Looking at the structures of each



team, common subblocks (e.g., differential amplifiers, current mirrors, etc.) are (re)used. We believe that this is an important aspect for automatic generation of analog circuits. In other words, analog circuits consist of a combination of subblocks with specific functions, which can be represented as shown in Figure 1.

We believe that the schematics and layouts of these blocks and

SOCIETY NEWS

SSCS PICO Chronicles: News From the Open Source Community

Code-a-Chip Travel Grant Awards at VLSI 2024

In the last edition of this column [1], we missed the message from one winning team (Figure 1), which we present in this edition.

Message From Tianyang Liu

It’s a great honor to have our work recognized by this prestigious competi-

the successful tape out of three different chips in the GlobalFoundries GF180MCU technology—of course, using open source IC design tools and the open source PDK. Teams from Pakistan (creating the chip named LTC1 [2]); Chile, Korea, and Japan (teaming up for LTC2 [3]); and

can be investigated and used as a starting point and inspiration for future designs. It is an excellent achievement that these teams could finalize their designs and proceed to a successful tape out—congratulations!

Everyone involved looks forward to receiving the

SSCS PICO Chronicles: Next Steps and Outreach

After the ground-breaking release of an open source production-ready 130-nm CMOS process design kit (PDK) (also known as SKY130) by Google and SkyWater Technologies in 2020, it did not take long for GlobalFoundries to follow suit and

The advent of open source IC design has transformed the global landscape, welcoming contributors from all corners of the world into the field of chip design and development. In Latin America, universities are joining this global endeavor, seizing

sat applications). The current Chipathon edition, centered on low-cost analog building block testing, features four candidate designs from Latin America: one from UFSC in Brazil, two from Universidad Técnica Federico Santa María (USM) in Chile,

what are **you** hoping to get
out of this?

overview: club structure



- ◆ weekly meetings for questions, feedback, etc
- ◆ regular workshops & speaker lectures
- ◆ mentoring and collaboration with faculty
- ◆ team-based or individual projects
- ◆ flexible time commitment

Next steps & upcoming milestone



- ◆ competition participation:
 - form at least one Brown team for IEEE Code-a-Chip 2026 (building on this year's winners)
 - Join next year's chipathon
- ◆ Host SSCS distinguished speakers
- ◆ Hold a hackathon/sprint on one of the open-source frameworks
- ◆ Club registration: finalize signatures (already mentioned on p. 23).
- ◆ Website/branding: launch a simple page under brown.edu/engineering or a GitHub Pages site to showcase projects.

looking forward



- ❖ organizing projects, speakers, and schedule over winter break.
- ❖ feel free to provide us feedback and ideas!

thank you for coming!

signatures for official club registration!

